

12.1 The impossible power

When alternating current systems were first introduced, learned scientists claimed that it was impossible to deliver energy by such a means. Their argument was that power transfer would take place during the first half of the cycle – and then it would transfer back during the second half.

Curiously there was some truth in what they claimed, but they had overlooked the basic relationship $p = i^2 R$. The square of the current means that the power is positive no matter whether the current has a positive or a negative value. But it is only the resistive element that dissipates energy from the circuit. Inductors and capacitors do not dissipate energy, which supports the theory of the impossible power.

Let us therefore examine in more detail the energy transfer process which takes place first in resistive circuits and then in reactive circuits.

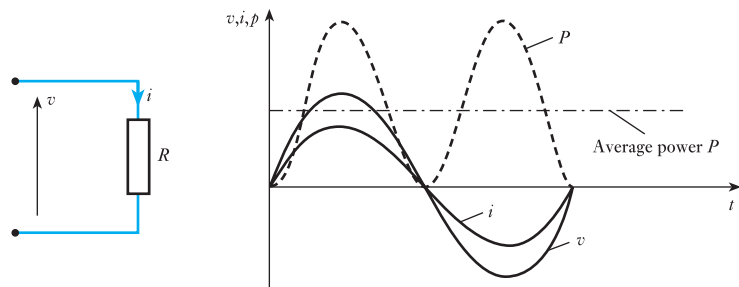
12.2 Power in a resistive circuit

In section 9.6 it was explained that when an alternating current flows through a resistor of R ohms, the average heating effect over a complete cycle is $I^2 R$ watts, where I is the r.m.s. value of the current in amperes.

If V volts is the r.m.s. value of the applied voltage, then for a non-reactive circuit having constant resistance R ohms, $V = IR$.

The waveform diagrams for resistance are shown in Fig. 12.1. To the current and voltage waves, there have been added the waves of the product vi . Since the instantaneous values of vi represent the instantaneous power p , it follows that these waves are the power waves. Because the power is continually fluctuating, the power in an a.c. circuit is taken to be the average value of the wave.

Fig. 12.1 Waveform diagrams for a resistive circuit



In the case of the pure resistance, the average power can be most easily obtained from the definition of the r.m.s. current in the circuit, i.e.

$$P = I^2 R \quad [12.1]$$

This relation can also be expressed as

$$P = VI \quad [12.2]$$

Hence the power in a non-reactive circuit is given by the product of the ammeter and voltmeter readings, exactly as in a d.c. circuit.

The power associated with energy transfer from the electrical system to another system such as heat, light or mechanical drives is termed active power, thus the average given by $I^2 R$ is the active power of the arrangement.

Alternatively, the average power can be derived from a formal analysis of the power waveform.

$$\begin{aligned}
 P &= \frac{\omega}{2\pi} \int_0^{\frac{2\pi}{\omega}} (V_m \sin \omega t \cdot I_m \sin \omega t) dt \\
 &= V_m I_m \frac{\omega}{2\pi} \int_0^{\frac{2\pi}{\omega}} (\sin^2 \omega t) dt \\
 &= V_m I_m \frac{\omega}{2\pi} \int_0^{\frac{2\pi}{\omega}} \left(\frac{1 - \cos 2\omega t}{2} \right) dt
 \end{aligned}$$

From this relation it can be seen that the wave has a frequency double that of the component voltage and current waves. This can be seen in Fig. 12.1; however, it also confirms that the wave is sinusoidal although it has been displaced from the horizontal axis.

$$\begin{aligned}
 P &= V_m I_m \frac{\omega}{2\pi} \left[\frac{t}{2} - \frac{\sin 2\omega t}{4\omega} \right]_0^{\frac{2\pi}{\omega}} \\
 &= V_m I_m \frac{\omega}{2\pi} \cdot \frac{2\pi}{2\omega}
 \end{aligned}$$

$$P = \frac{V_m I_m}{2} \quad [12.3]$$

$$P = VI$$

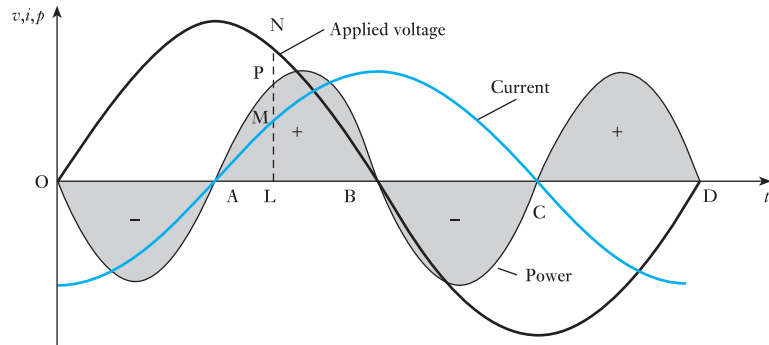
12.3

Power in a purely inductive circuit

Consider a coil wound with such thick wire that the resistance is negligible in comparison with the inductive reactance X_L ohms. If such a coil is connected across a supply voltage V , the current is given by $I = V/X_L$ amperes. Since the resistance is very small, the heating effect and therefore the active power are also very small, even though the voltage and the current are large. Such a curious conclusion – so different from anything we have experienced in d.c. circuits – requires fuller explanation if its significance is to be properly understood. Let us therefore consider Fig. 12.2, which shows the applied voltage and the current for a purely inductive circuit, the current lagging the voltage by a quarter of a cycle.

The power at any instant is given by the product of the voltage and the current at that instant; thus at instant L, the applied voltage is LN volts and the current is LM amperes, so that the power at that instant is LN $\times$ LM watts and is represented to scale by LP.

Fig. 12.2 Power curve for a purely inductive circuit



By repeating this calculation at various instants we can deduce the curve representing the variation of power over one cycle. It is seen that during interval OA the applied voltage is positive, but the current is negative, so that the power is negative; and that during interval AB , both the current and the voltage are positive, so that the power is positive.

The power curve is found to be symmetrical about the horizontal axis OD . Consequently the shaded areas marked '−' are exactly equal to those marked '+', so that the mean value of the power over the complete cycle OD is zero.

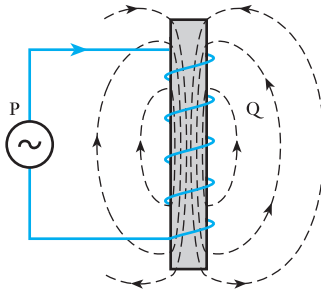


Fig. 12.3 Magnetic field of an inductive circuit

It is necessary, however, to consider the significance of the positive and negative areas if we are to understand what is really taking place. So let us consider an a.c. generator P (Fig. 12.3) connected to a coil Q whose resistance is negligible, and let us assume that the voltage and current are represented by the graphs in Fig. 12.2. At instant A , there is no current and therefore no magnetic field through and around Q . During interval AB , the growth of the current is accompanied by a growth of flux as shown by the dotted lines in Fig. 12.3. But the existence of a magnetic field involves some kind of a strain in the space occupied by the field and the storing up of energy in that field, as already dealt with in section 8.10. The current, and therefore the magnetic energy associated with it, reach their maximum values at instant B ; and, since the loss in the coil is assumed negligible, it follows that at that instant the whole of the energy supplied to the coil during interval AB , and represented by the shaded area marked '+', is stored up in the magnetic field.

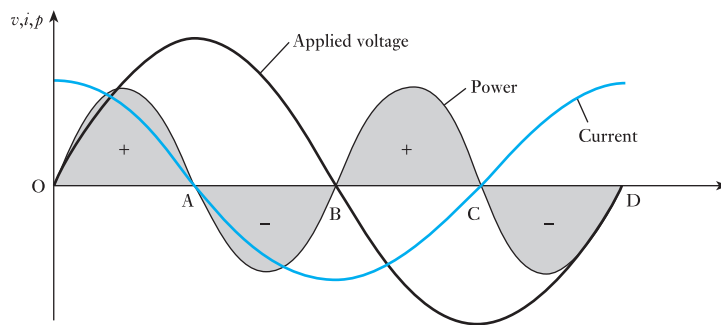
During the interval BC the current and its magnetic field are decreasing; and the e.m.f. induced by the collapse of the magnetic flux is in the same direction as the current. But any circuit in which the current and the induced or generated e.m.f. are in the same direction acts as a source of electrical energy (see section 8.2). Consequently the coil is now acting as a generator transforming the energy of its magnetic field into electrical energy, the latter being sent to generator P to drive it as a motor. The energy thus returned is represented by the shaded area marked '−' in Fig. 12.2; and since the positive and negative areas are equal, it follows that during alternate quarter-cycles electrical energy is being sent from the generator to the coil, and during the other quarter-cycles the same amount of energy is sent back from the coil to the generator. Consequently the net energy absorbed by the coil during a

complete cycle is zero; in other words, the average power over a complete cycle is zero.

12.4 Power in a purely capacitive circuit

In this case, the current leads the applied voltage by a quarter of a cycle, as shown in Fig. 12.4; and by multiplying the corresponding instantaneous values of the voltage and current, we can derive the curve representing the variation of power. During interval OA, the voltage and current are both positive so that the power is positive, i.e. power is being supplied from the generator to the capacitor, and the shaded area enclosed by the power curve during interval OA represents the value of the electrostatic energy stored in the capacitor at instant A.

Fig. 12.4 Power curve for a purely capacitive circuit



During interval AB, the p.d. across the capacitor decreases from its maximum value to zero and the whole of the energy stored in the capacitor at instant A is returned to the generator; consequently the net energy absorbed during the half-cycle OB is zero. Similarly, the energy absorbed by the capacitor during interval BC is returned to the generator during interval CD. Hence the average power over a complete cycle is zero.

That the inductive and capacitive circuits do not dissipate power can be proved by an analysis of the power wave. Consider the case of the capacitor.

$$\begin{aligned}
 P &= \frac{\omega}{2\pi} \int_0^{\frac{2\pi}{\omega}} (V_m \sin \omega t \cdot I_m \cos \omega t) dt \\
 &= V_m I_m \frac{\omega}{2\pi} \int_0^{\frac{2\pi}{\omega}} (\sin \omega t \cdot \cos \omega t) dt \\
 &= V_m I_m \frac{\omega}{2\pi} \int_0^{\frac{2\pi}{\omega}} \frac{\sin 2\omega t}{2} \cdot dt \\
 &= V_m I_m \frac{\omega}{2\pi} \left[-\frac{\cos 2\omega t}{4\omega} \right]_0^{\frac{2\pi}{\omega}}
 \end{aligned}$$

$$P = 0$$

[12.4]

12.5

Power in a circuit with resistance and reactance

Let us consider the general case of the current differing in phase from the applied voltage; thus in Fig. 12.5(a), the current is shown lagging the voltage by an angle ϕ .

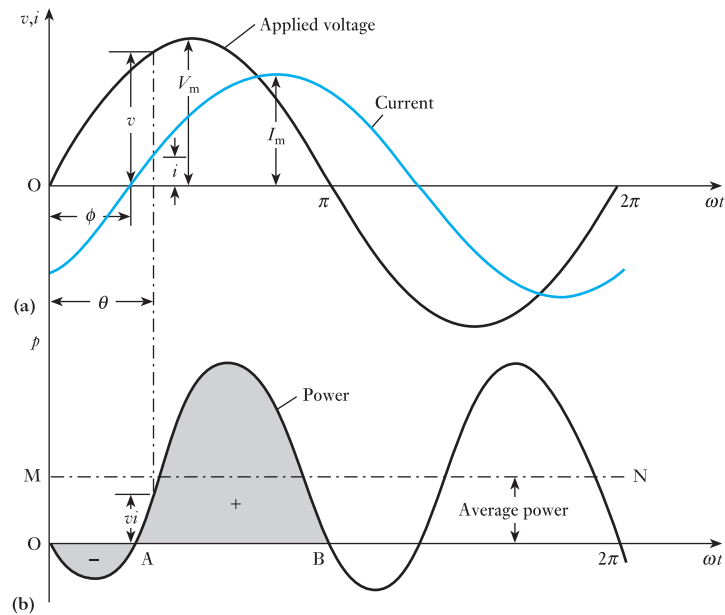
Let instantaneous value of voltage be

$$v = V_m \sin \omega t$$

then instantaneous value of current is

$$i = I_m \sin(\omega t - \phi)$$

Fig. 12.5 Voltage, current and power curves



At any instant, the value of the power is given by the product of the voltage and the current at that instant, i.e. instantaneous value of power = vi watts.

By multiplying the corresponding instantaneous values of voltage and current, the curve representing the variation of power in Fig. 12.5(b) can be derived, i.e. instantaneous power is

$$\begin{aligned} vi &= V_m \sin \omega t \cdot I_m \sin(\omega t - \phi) \\ &= \frac{1}{2} V_m I_m \{ \cos \phi - \cos(2\omega t - \phi) \} \\ &= \frac{1}{2} V_m I_m \cos \phi - \frac{1}{2} V_m I_m \cos(2\omega t - \phi) \end{aligned}$$

From this expression, it is seen that the instantaneous value of the power consists of two components:

1. $\frac{1}{2} V_m I_m \cos \phi$, which contains no reference to ωt and therefore remains constant in value.
2. $\frac{1}{2} V_m I_m \cos(2\omega t - \phi)$, the term $2\omega t$ indicating that it varies at twice the supply frequency; thus in Fig. 12.5(b) it is seen that the power undergoes two cycles of variation for one cycle of the voltage wave. Furthermore, since the average value of a cosine curve over a *complete* cycle is zero, it

follows that this component does not contribute anything towards the *average* value of the power taken from the generator.

Hence, average power over one cycle is

$$\frac{1}{2} V_m I_m \cos \phi = \frac{V_m}{\sqrt{2}} \cdot \frac{I_m}{\sqrt{2}} \cdot \cos \phi$$

$$\therefore P = VI \cos \phi \quad [12.5]$$

where V and I are the r.m.s. values of the voltage and current respectively. In Fig. 12.5(b), the average power is represented by the height above the horizontal axis of the dotted line MN drawn midway between the positive and negative peaks of the power curve.

It will be noticed that during interval OA in Fig. 12.5(b), the power is negative, and the shaded negative area represents energy returned from the circuit to the generator. The shaded positive area during interval AB represents energy supplied from the generator to the circuit, and the difference between the two areas represents the net energy absorbed by the circuit during interval OB. The larger the phase difference between the voltage and current, the smaller is the difference between the positive and negative areas and the smaller, therefore, is the average power over the complete cycle.

The average power over the complete cycle is the active power, which is measured in watts.

The product of the voltage and the current in an a.c. circuit is termed the apparent power.

Apparent power Symbol: S Unit: **voltampere (V A)**

$$S = VI \quad [12.6]$$

$$\therefore P = VI \cos \phi$$

$$\text{and } P = S \cos \phi \quad [12.7]$$

Example 12.1

A coil having a resistance of 6Ω and an inductance of 0.03 H is connected across a 50 V , 60 Hz supply. Calculate:

- the current;
- the phase angle between the current and the applied voltage;
- the apparent power;
- the active power.

(a) The phasor diagram for such a circuit is given in Fig. 12.6.

$$\begin{aligned} \text{Reactance of circuit} &= 2\pi fL = 2 \times 3.14 \times 60 \times 0.03 \\ &= 11.31 \Omega \end{aligned}$$

From equation [10.10]

$$\text{Impedance} = \sqrt{6^2 + (11.31)^2} = 12.8 \Omega$$

$$\text{and Current} = \frac{50}{12.8} = 3.9 \text{ A}$$

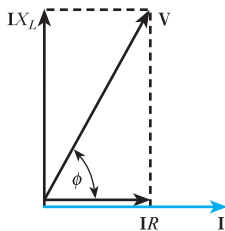


Fig. 12.6 Phasor diagram for Example 12.1

(b) From equation [10.11]

$$\tan \phi = \frac{X}{R} = \frac{11.31}{6} = 1.885$$

$$\phi = 62^\circ 3'$$

(c) Apparent power $S = 50 \times 3.91 = 196 \text{ V A}$

(d) Active power = apparent power $\times \cos \phi$
 $= 195.5 \times 0.469 = 92 \text{ W}$

Alternatively:

$$\text{Active power} = I^2 R = (3.91)^2 \times 6 = 92 \text{ W}$$

Example 12.2

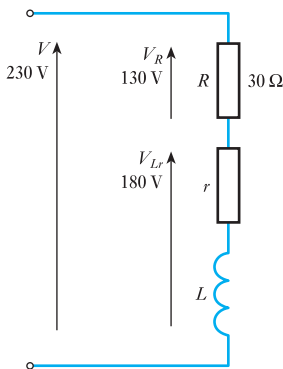


Fig. 12.7 Circuit diagram for Example 12.2

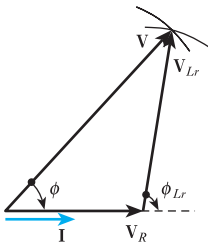


Fig. 12.8 Phasor diagram for Example 12.2

An inductor coil is connected in series with a pure resistor of 30Ω across a 230 V , 50 Hz supply. The voltage measured across the coil is 180 V and the voltage measured across the resistor is 130 V (Fig. 12.7). Calculate the power dissipated in the coil.

The complexor diagram is constructed by first drawing the complexor I (Fig. 12.8). The resistor voltage complexor V_R is then drawn in phase with I . Since neither the coil phase angle nor the circuit phase angle are known, it is necessary to derive the remainder of the diagram by construction to scale. Circles of radius V and V_{Lr} are drawn radiating from the appropriate ends of V_R . The point of intersection of the circles satisfies the relation

$$V = V_R + V_{Lr} \quad (\text{phasor sum})$$

By the geometry of the diagram:

$$V^2 = V_R^2 + V_{Lr}^2 + 2V_R V_{Lr} \cos \phi_{Lr}$$

$$230^2 = 130^2 + 180^2 + 2 \times 130 \times 180 \times \cos \phi_{Lr}$$

$$\cos \phi_{Lr} = 0.077 \text{ lag}$$

$$I = \frac{V_R}{R} = \frac{130}{30} = 4.33 \text{ A}$$

$$P_r = V_{Lr} I \cos \phi_{Lr} = 180 \times 4.33 \times 0.077 = 60 \text{ W}$$

Alternatively:

$$Z_{Lr} = \frac{V_{Lr}}{I} = \frac{180}{4.33} = 41.5 \Omega$$

$$r = Z_{Lr} \cos \phi_{Lr} = 41.5 \times 0.077 = 3.20 \Omega$$

$$P_r = I^2 r = 4.33^2 \times 3.20 = 60 \text{ W}$$

12.6

Power factor

In a.c. work, the product of the r.m.s. values of the applied voltage and current is VI . It has already been shown that the active power $P = VI \cos \phi$ and the value of $\cos \phi$ has to lie between 0 and 1. It follows that the active power P can be either equal to or less than the product VI , which is termed the apparent power and is measured in voltamperes (VA).

The ratio of the active power P to the apparent power S is termed the power factor, i.e.

$$\frac{\text{Active power } P \text{ in watts}}{\text{Apparent power } S \text{ in voltamperes}} = \text{power factor}$$

$$\therefore \cos \phi = \frac{P}{S} = \frac{P}{VI} \quad [12.8]$$

$$\text{or} \quad \text{Active power } P = \text{apparent power } S \times \text{power factor} \quad [12.9]$$

Comparison of expressions [12.5] and [12.8] shows that for *sinusoidal* voltage and current:

$$\text{Power factor} = \cos \phi$$

From the general phasor diagram of Fig. 10.26 for a *series* circuit, it follows that

$$\cos \phi = \frac{IR}{V} = \frac{IR}{IZ} = \frac{\text{resistance}}{\text{impedance}}$$

$$\therefore \cos \phi = \frac{R}{Z} \quad [12.10]$$

It has become the practice to say that the power factor is *lagging* when the *current lags the supply voltage*, and *leading* when the *current leads the supply voltage*. This means that the supply voltage is regarded as the reference quantity.

Example 12.3

An inductor coil is connected to a supply of 230 V at 50 Hz and takes a current of 5.0 A. The coil dissipates 750 W (Fig. 12.9). Calculate:

- the resistance and the inductance of the coil;
- the power factor of the coil.

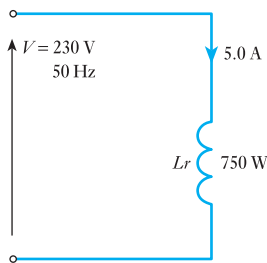


Fig. 12.9 Circuit diagram for Example 12.3

In this example, the symbol r will be used to denote the resistance of the coil instead of R . This is done to draw attention to the fact that the resistance is not a separate component of the circuit but is an integral part of the inductor coil. This device was also used in Example 12.2.

$$(a) \quad Z = \frac{V}{I} = \frac{230}{5} = 46 \, \Omega$$

$$r = \frac{P}{I^2} = \frac{750}{5^2} = 30 \, \Omega$$

$$X_L = (Z^2 - r^2)^{\frac{1}{2}} = (46^2 - 30^2)^{\frac{1}{2}} = 34.87 \, \Omega$$

$$L = \frac{X_L}{2\pi f} = \frac{34.87}{2\pi 50} = \frac{34.87}{314} = 0.111 \, \text{H} = 111 \, \text{mH}$$

$$(b) \quad \text{Power factor} = \cos \phi = \frac{P}{S} = \frac{R}{VI} = \frac{750}{230 \times 5} = 0.65 \, \text{lag}$$

12.7 Active and reactive currents

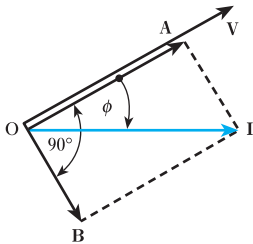


Fig. 12.10 Active and reactive components of current

If a current I lags the applied voltage V by an angle ϕ , as in Fig. 12.10, it can be resolved into two components, OA in phase with the voltage and OB lagging by 90° .

If the phasor diagram in Fig. 12.10 refers to a circuit possessing resistance and inductance in series, OA and OB must not be labelled I_R and I_L respectively. Such terms should only be applied to branch currents as would be the case if R and L were in parallel. This error of applying parallel terms to series circuits is easily made by beginners – **you have been warned!**

Since

$$\text{Power} = IV \cos \phi = V \times OI \cos \phi = V \times OA \text{ watts}$$

therefore OA is termed the *active* component of the current, i.e.

$$\text{Active component of current} = I \cos \phi \quad [12.11]$$

Power due to component OB is

$$V \times OB \cos 90^\circ = 0$$

so that OB is termed the *reactive* component of the current, i.e.

$$\text{Reactive component of current} = I \sin \phi \quad [12.12]$$

and Reactive power Q in vars = $VI \sin \phi$

The term ‘var’ is short for voltampere reactive.

$$\therefore Q = VI \sin \phi \quad [12.13]$$

$$\begin{aligned} \text{also } P^2 + Q^2 &= (VI \cos \phi)^2 + (VI \sin \phi)^2 \\ &= (VI)^2 (\cos^2 \phi + \sin^2 \phi) = (VI)^2 = S^2 \end{aligned}$$

$$S^2 = P^2 + Q^2 \quad [12.14]$$

Example 12.4

A single-phase motor operating off a 400 V, 50 Hz supply is developing 10 kW with an efficiency of 84 per cent and a power factor (p.f.) of 0.7 lagging. Calculate:

- the input apparent power;
- the active and reactive components of the current;
- the reactive power (in kilovars).

$$\begin{aligned} \text{(a) Efficiency} &= \frac{\text{output power in watts}}{\text{input power in watts}} \\ &= \frac{\text{output power in watts}}{IV \times \text{p.f.}} \end{aligned}$$

$$\therefore 0.84 = \frac{10 \times 1000}{IV \times 0.7}$$

$$\text{so that } IV = 17\,000 \text{ V A}$$

$$\therefore \text{Input} = 17.0 \text{ kVA}$$

$$\begin{aligned}
 \text{(b) Current taken by motor} &= \frac{\text{input volt amperes}}{\text{voltage}} \\
 &= \frac{17\,000}{400} = 42.5 \text{ A}
 \end{aligned}$$

therefore active component of current is

$$42.5 \times 0.7 = 29.75 \text{ A}$$

Since

$$\begin{aligned}
 \sin \phi &= \sqrt{1 - \cos^2 \phi} = \sqrt{1 - (0.7)^2} \\
 &= 0.714
 \end{aligned}$$

therefore reactive component of current is

$$42.5 \times 0.714 = 30.4 \text{ A}$$

$$\begin{aligned}
 \text{(c) Reactive power} &= 400 \times \frac{30.35}{1000} \\
 &= 12.1 \text{ kvar}
 \end{aligned}$$

Example 12.5

Calculate the capacitance required in parallel with the motor of Example 12.4 to raise the supply power factor to 0.9 lagging.

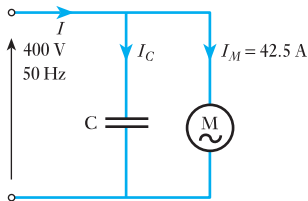


Fig. 12.11 Circuit diagram for Example 12.5

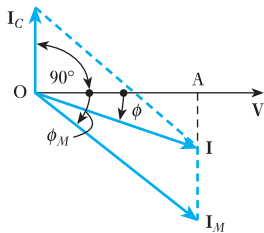


Fig. 12.12 Phasor diagram for Fig. 12.11

The circuit and phasor diagrams are given in Figs 12.11 and 12.12 respectively, M being the motor taking a current I_M of 42.5 A.

Current I_C taken by the capacitor must be such that when combined with I_M , the resultant current I lags the voltage by an angle ϕ , where $\cos \phi = 0.9$. From Fig. 12.12,

$$\begin{aligned}
 \text{Active component of } I_M &= I_M \cos \phi_M \\
 &= 42.5 \times 0.7 \\
 &= 29.75 \text{ A}
 \end{aligned}$$

and active component of I is

$$I \cos \phi = I \times 0.9$$

These components are represented by OA in Fig. 12.12.

$$\therefore I = \frac{29.75}{0.9} = 33.06 \text{ A}$$

$$\begin{aligned}
 \text{Reactive component of } I_M &= I_M \sin \phi_M \\
 &= 30.35 \text{ A (from Example 12.4)}
 \end{aligned}$$

and reactive component of I is

$$\begin{aligned}
 I \sin \phi &= 33.06 \sqrt{1 - (0.9)^2} \\
 &= 33.06 \times 0.436 \\
 &= 14.4 \text{ A}
 \end{aligned}$$

From Fig. 12.12 it will be seen that

$$\begin{aligned} I_C &= \text{reactive component of } I_M - \text{reactive component of } I \\ &= 30.35 - 14.4 \\ &= 15.95 \text{ A} \end{aligned}$$

But $I_C = 2\pi fCV$

$$\therefore 15.95 = 2 \times 3.14 \times 50 \times C \times 400$$

and $C = 127 \times 10^{-6} \text{ F} = 127 \mu\text{F}$

From Example 12.5 it will be seen that the effect of connecting a $127 \mu\text{F}$ capacitor in parallel with the motor is to reduce the current taken from the supply from 42.5 to 33.1 A, without altering either the current or the power taken by the motor. This enables an economy to be effected in the size of the generating plant and in the cross-sectional area of conductor in the supply cable.

Example 12.6

An a.c. generator is supplying a load of 300 kW at a power factor of 0.6 lagging. If the power factor is raised to unity, how much more power (in kilowatts) can the generator supply for the same kilovolt-ampere loading?

Since the power in kW is number of kilovoltamperes $\times$ power factor, therefore number of kilovoltamperes is

$$\frac{300}{0.6} = 500 \text{ kVA}$$

When the power factor is raised to unity:

$$\text{Number of kilowatts} = \text{number of kilovoltamperes} = 500 \text{ kW}$$

Hence increased power supplied by generator is

$$500 - 300 = 200 \text{ kW}$$

12.8

The practical importance of power factor

If an a.c. generator is rated to give, say, 2000 A at a voltage of 400 V, it means that these are the highest current and voltage values the machine can give without the temperature exceeding a safe value. Consequently the rating of the generator is given as $400 \times 2000/1000 = 800 \text{ kVA}$. The phase difference between the voltage and the current depends upon the nature of the load and not upon the generator. Thus if the power factor of the load is unity, the 800 kVA are also 800 kW, and the engine driving the generator has to be capable of developing this power together with the losses in the generator. But if the power factor of the load is, say, 0.5, the power is only 400 kW, so that the engine is developing only about one-half of the power of which it is capable, though the generator is supplying its rated output of 800 kVA.

Similarly, the conductors connecting the generator to the load have to be capable of carrying 2000 A without excessive temperature rise. Consequently they can transmit 800 kW if the power factor is unity, but only 400 kW at 0.5 power factor for the same rise of temperature.

20.1 Rectifier circuits

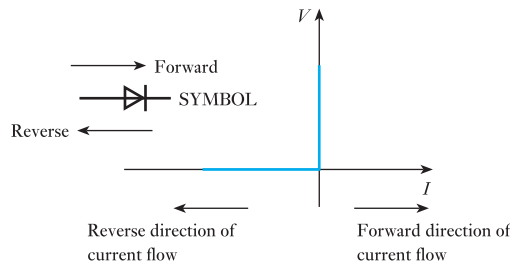
Since a diode has the characteristic of having a much greater conductivity in one direction than in the other, it will produce a direct component of current when connected in series with an alternating voltage and a load. This process is known as rectification and is the main use to which diodes are put. There are numerous applications for rectification, e.g. driving a d.c. motor from a.c. mains and the production of direct-voltage supplies for electronic amplifiers.

20.2 Half-wave rectifier

While alternating currents and voltages play the leading roles in most electrical and electronic equipment, nevertheless many devices can either only operate on unidirectional currents and voltages, or at least they require such a supply as part of their mode of operation. The process of obtaining unidirectional currents and voltages from alternating currents and voltages is termed *rectification*.

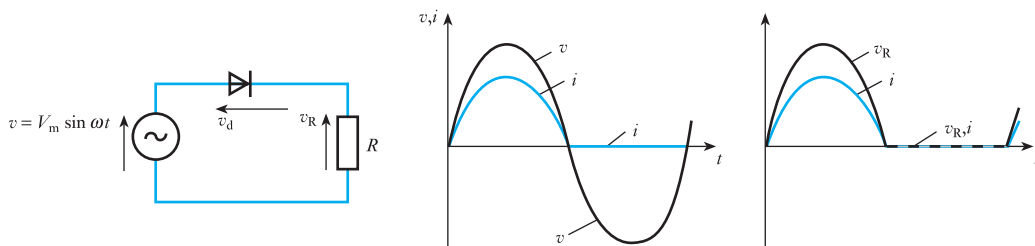
The device that makes such a process possible is a diode, the ideal operating characteristic of which is given in Fig. 20.1. When the applied voltage acts in the forward direction, there is no voltage drop across the diode and a current flows unimpeded. However, when the applied voltage acts in the reverse direction, a voltage drop appears across the diode and no current flows.

Fig. 20.1 Ideal diode characteristics



It is possible to obtain rectification by means of a single diode as indicated in Fig. 20.2. The current can only flow through the diode in one direction and thus the load current can only flow during alternate half-cycles. For this reason, the system is known as half-wave rectification. The load current, and hence the voltage drop across the load, is unidirectional and could be described as direct, although this term is more usually reserved for steady unidirectional quantities.

Fig. 20.2 Half-wave rectification



With reference to the circuit shown in Fig. 20.2:

v = instantaneous supply voltage

v_d = instantaneous voltage across the diode

v_R = instantaneous voltage across the load resistance

i = instantaneous diode current

Using Kirchhoff's second law in the closed loop:

$$v = v_d + v_R = v_d + iR$$

$$i = -\frac{1}{R}v_d + \frac{v}{R} \quad [20.1]$$

Examination of the characteristics shown in Figs 19.17 and 19.18 indicates that v_d has a small value during the positive half-cycle when the diode conducts, but is equal to v during the negative half-cycle, there being negligible current and v_R consequently also being negligible. However, if we simplify the voltage/current characteristic to the idealized one shown in Fig. 20.1, then the value of v_d is zero and the effective resistance r_d (termed the *forward resistance*) of the diode is zero. If the supply voltage is

$$v = V_m \sin \omega t$$

$$\text{then } i = \frac{v}{R} = \frac{V_m \sin \omega t}{R} = I_m \sin \omega t \quad \text{during the positive half-cycle}$$

Thus if the mean value of the current (neglecting the reverse current) = I_{dc} then:

$$\begin{aligned} I_{dc} &= \frac{1}{2\pi} \int_0^\pi I_m \sin \omega t \, d(\omega t) = \frac{I_m}{2\pi} [-\cos \omega t]_0^\pi \\ &= \frac{I_m}{2\pi} [-\cos \pi + \cos 0] = \frac{I_m}{2\pi} [1 + 1] \end{aligned}$$

$$\therefore I_{dc} = \frac{I_m}{\pi} = 0.318 I_m \quad [20.2]$$

Similarly if the r.m.s. value of the current is I_{rms} then:

$$\begin{aligned} I_{rms} &= \sqrt{\left[\frac{1}{2\pi} \int_0^\pi I_m^2 \sin^2 \omega t \, d(\omega t) \right]} \\ &= \sqrt{\left[\frac{I_m^2}{2\pi} \int_0^\pi \frac{1}{2} (1 - \cos 2\omega t) \, d(\omega t) \right]} \\ &= \sqrt{\left\{ \frac{I_m^2}{2\pi} \times \frac{1}{2} \left[\omega t + \frac{1}{2} \sin 2\omega t \right]_0^\pi \right\}} = \sqrt{\left\{ \frac{I_m^2}{4\pi} [\pi] \right\}} \end{aligned}$$

$$\therefore I_{rms} = \frac{I_m}{2} = 0.5 I_m \quad [20.3]$$

The voltage across the load during the positive half-cycles is given by

$$v_R = Ri = RI_m \sin \omega t = V_m \sin \omega t$$

Therefore, in the same way as derived for the current:

$$\text{Mean value of the load voltage} = V_{dc} = \frac{V_m}{\pi} \quad [20.4]$$

$$\text{RMS value of the load voltage} = V_{rms} = \frac{V_m}{2} \quad [20.5]$$

The maximum voltage, which occurs across the diode in the reverse direction, is known as the peak inverse voltage (PIV). This must be less than the breakdown voltage of the diode if it is not to conduct appreciably in the reverse direction. The peak inverse voltage for the diode in this circuit occurs when the potential of B is positive with respect to A by its maximum amount. The reverse resistance of the diode will, in the great majority of practical cases, be very much greater than the load resistance and most of the applied voltage will appear across the diode. Thus the peak inverse voltage equals approximately the peak value of the supply voltage.

Since the production of direct current from an a.c. supply is the object of the circuit, the useful power output is that produced in the load by the d.c. component of the load current. The efficiency of a rectifier circuit is defined as

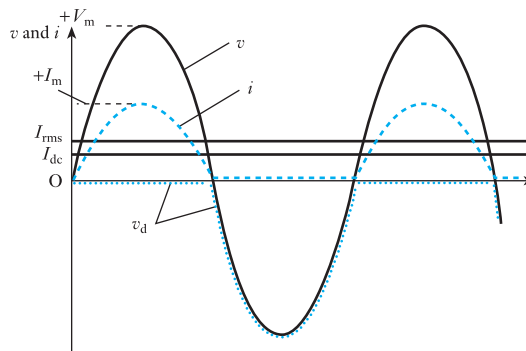
$$\begin{aligned} \text{Efficiency } \eta &= \frac{\text{power in the load due to d.c. component of current}}{\text{total power dissipated in the circuit}} \\ &= \frac{I_{dc}^2 R}{I_{rms}^2 R} = \left[\frac{I_m}{\pi} \right]^2 \times \left[\frac{2}{I_m} \right]^2 = 0.405 \end{aligned}$$

This efficiency is based on $r_d = 0$. If it had a greater value, the efficiency would be reduced since the total power would be $I_{rms}^2(r_d + R)$, hence

$$\eta_m = 0.405 \quad [20.6]$$

Waveforms for the half-wave rectifier circuit are shown in Fig. 20.3.

Fig. 20.3 Waveforms for half-wave rectifier circuit



20.3 Full-wave rectifier network

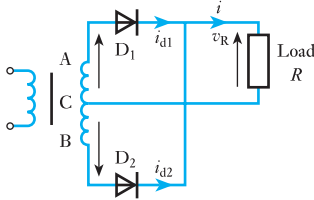


Fig. 20.4 Full-wave rectifier with resistive load

The half-wave rectifier gave rise to an output which had a unidirectional current, but the resulting current does not compare favourably with the direct current that one would expect from, say, a battery. One reason is that the half-wave rectifier is not making use of the other half of the supply waveform. It would be technically and economically advantageous if both halves were rectified and this may be achieved by a full-wave rectifier network.

The basic full-wave rectifier network is shown in Fig. 20.4. Here C is a centre tap on the secondary of the transformer, thus the e.m.f.s induced in each section of the secondary are equal, and when the potential of A is positive with respect to C, so is that of C positive with respect to B. With these polarities, diode D_1 will conduct while diode D_2 is non-conducting. When these polarities reverse, diode D_2 will conduct and D_1 will be non-conducting. In this way each diode conducts on alternate half-cycles, passing current through the load in the same direction. For

$$v_{AC} = v = V_m \sin \omega t$$

$$v_{BC} = -v = -V_m \sin \omega t$$

For identical diodes with a forward resistance $r_d = 0$ and an infinite reverse resistance, then during the period that v is positive:

$$\text{Diode 1 current} = i_{d1} = \frac{V_m \sin \omega t}{R} = I_m \sin \omega t$$

$$\text{Diode 2 current} = i_{d2} = 0$$

During the period that v is negative:

$$\text{Diode 1 current} = i_{d1} = 0$$

$$\text{Diode 2 current} = i_{d2} = \frac{-V_m \sin \omega t}{R} = -I_m \sin \omega t$$

At any instant the load current is given by

$$i = i_{d1} + i_{d2}$$

Thus in this circuit the current will repeat itself twice every cycle of the supply voltage, therefore mean value of the load current is

$$I_{dc} = \frac{1}{\pi} \int_0^\pi I_m \sin \omega t \, d(\omega t)$$

$$I_{dc} = \frac{2I_m}{\pi} = 0.637 I_m \quad [20.7]$$

RMS value of the load current is

$$I_{rms} = \sqrt{\left[\frac{1}{\pi} \int_0^\pi I_m^2 \sin^2 \omega t \, d(\omega t) \right]}$$

$$I_{rms} = \frac{I_m}{\sqrt{2}} = 0.707 I_m \quad [20.8]$$

Similarly, for the load voltage, mean value of the load voltage is

$$V_{dc} = \frac{2V_m}{\pi} \quad [20.9]$$

RMS value of the load voltage is

$$V_{rms} = \frac{V_m}{\sqrt{2}} \quad [20.10]$$

For this circuit

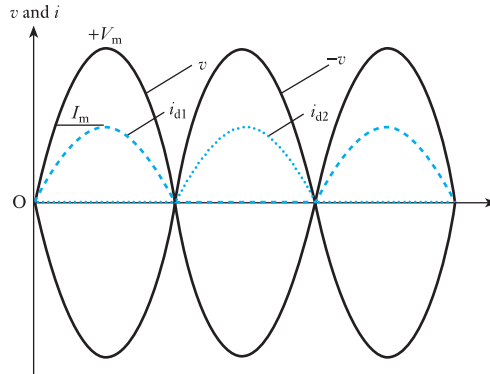
$$\eta = \frac{I_{dc}^2 R}{I_{rms}^2 R} = \left[\frac{2I_m}{\pi} \right]^2 \times \left[\frac{\sqrt{2}}{I_m} \right]^2 = \frac{8}{\pi^2}$$

Again, since $r_d = 0$,

$$\eta = 0.81 \quad [20.11]$$

Waveforms for the full-wave rectifier network are shown in Fig. 20.5.

Fig. 20.5 Waveforms for full-wave rectifier circuit

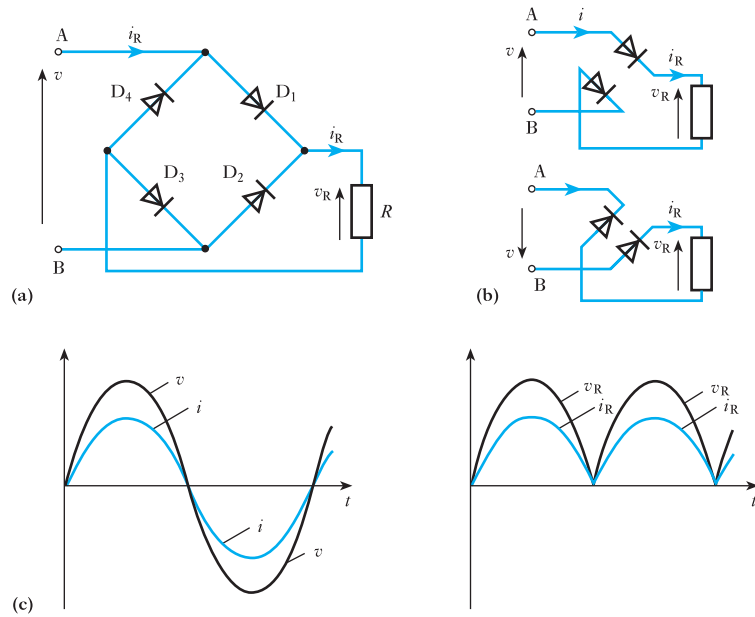


20.4 Bridge rectifier network

The full-wave rectifier only makes use of each half of the transformer winding for half of the time. The winding can be used all of the time, or the transformer can be omitted in certain cases by the application of the bridge rectifier network. Such a network is shown in Fig. 20.6(a).

When the potential of A is positive with respect to B, diodes D_1 and D_3 conduct and a current flows in the load. When the potential of B is positive with respect to A, diodes D_2 and D_4 conduct and the current in the load is in the same direction as before. Thus a full-wave type of output is obtained. The expressions derived for current and load voltage for the full-wave circuit will be applicable here also. Note should be taken, however, that, at any instant, two diodes are conducting: Fig. 20.6(b) shows the active circuits during each half-cycle.

The waveform diagrams shown in Fig. 20.6(c) indicate that the current i taken from the supply is purely alternating yet the load current i_R is unidirectional and therefore basically a direct current.

Fig. 20.6 Full-wave bridge rectification

The efficiency of the network is given by

$$\eta = \frac{I_{dc}^2 R}{I_{rms}^2 R} = \left[\frac{2}{\pi} I_m \right]^2 \times \left[\frac{\sqrt{2}}{I_m} \right]^2 = \frac{8}{\pi^2} \quad [20.12]$$

Neglecting the forward resistances of the diodes, the voltage which appears across the non-conducting diodes is equal to the supply voltage. Thus the peak inverse voltage is equal to the peak value of the supply voltage.

There are certain advantages to be obtained by using the bridge circuit as opposed to the centre-tapped secondary full-wave circuit. For a given mean load voltage, the bridge circuit uses only half the number of secondary turns, and, for a diode of given maximum peak-inverse-voltage rating, twice the mean output voltage can be obtained from it since the peak inverse voltage encountered is V_m compared with $2V_m$ for the full-wave circuit.

Example 20.1

The four diodes used in a bridge rectifier circuit have forward resistances which may be considered negligible, and infinite reverse resistances. The alternating supply voltage is 230 V r.m.s. and the resistive load is 46.0 Ω . Calculate:

- the mean load current;
- the rectifier efficiency.

$$(a) \quad I_m = \frac{\sqrt{2} \times 230}{46} = 7.07 \text{ A}$$

$$I_{dc} = \frac{2I_m}{\pi} = \frac{2 \times 7.07}{\pi} = 4.5 \text{ A}$$

$$(b) \quad I_{\text{rms}} = \frac{I_m}{\sqrt{2}} = \frac{7.07}{\sqrt{2}} = 5.0 \text{ A}$$

$$\therefore \quad \eta = \frac{4.5^2 \times 48}{5.0^2 \times 48} = 0.81 \quad (\text{i.e. 81 per cent})$$

Example 20.2

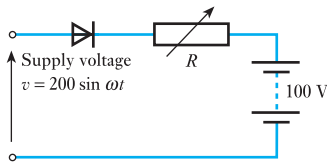


Fig. 20.7 Circuit for Example 20.2

A battery-charging circuit is shown in Fig. 20.7. The forward resistance of the diode can be considered negligible and the reverse resistance infinite. The internal resistance of the battery is negligible. Calculate the necessary value of the variable resistance R so that the battery charging current is 1.0 A.

Diode conducts during the period $v > 100 \text{ V}$.

$$200 \sin \omega t = 100$$

$$\text{i.e.} \quad \sin \omega t = 0.5$$

i.e. when

$$\omega t = \frac{\pi}{6} \quad \text{and} \quad \frac{5\pi}{6}$$

Therefore diode conducts when

$$\frac{\pi}{6} < \omega t < \frac{5\pi}{6}$$

During conduction

$$i = \frac{v - 100}{R} = \frac{200 \sin \omega t - 100}{R}$$

Mean value of current is

$$\begin{aligned} 1.0 &= \frac{1}{2\pi} \int_{\pi/6}^{5\pi/6} \frac{200 \sin \omega t - 100}{R} d(\omega t) \\ &= \frac{1}{2\pi R} \left[-200 \cos \omega t - 100\omega t \right]_{\pi/6}^{5\pi/6} \\ &= \frac{1}{2\pi R} \left[-200 \cos \frac{5\pi}{6} - 100 \times \frac{5\pi}{6} + 200 \cos \frac{\pi}{6} + 100 \times \frac{\pi}{6} \right] \\ &= \frac{1}{2\pi R} \left[-200 \left(-\frac{\sqrt{3}}{2} \right) - \frac{500\pi}{6} + 200 \left(\frac{\sqrt{3}}{2} \right) + \frac{100\pi}{6} \right] \\ &= \frac{1}{2\pi R} \left[200\sqrt{3} - \frac{400\pi}{6} \right] \\ R &= \frac{200\sqrt{3}}{2\pi} - \frac{400\pi}{2\pi \times 6} \\ R &= 55.1 - 33.3 = 21.8 \, \Omega \end{aligned}$$

20.5

Smoothing

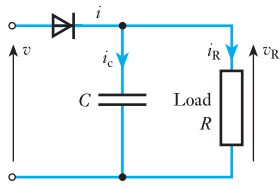


Fig. 20.8 Half-wave rectifier with capacitor input filter

The rectifier circuits so far described have produced, as required, a direct component of current in the load. There remains, however, a large alternating component. In a large number of applications it is desirable to keep this latter component small. This can be accomplished by the use of smoothing circuits, the simplest of which consists of a capacitor in parallel with the load. Figure 20.8 shows such an arrangement.

The diode conducts when the supply voltage v is more positive than the load voltage v_R . During this conduction period, if the diode forward resistance is neglected, then the load voltage is equal to the supply voltage. Therefore if

$$v = V_m \sin \omega t$$

$$i_R = \frac{v}{R} = \frac{V_m \sin \omega t}{R}$$

$$i_C = C \frac{dv}{dt} = C\omega V_m \cos \omega t$$

$$\text{Diode current} = i = i_R + i_C = \frac{V_m}{R} \sin \omega t + C\omega V_m \cos \omega t$$

$$= \sqrt{\left[\left(\frac{V_m}{R}\right)^2 + (\omega C V_m)^2 \cdot \sin(\omega t + \phi)\right]}$$

$$\therefore i = \frac{V_m}{R} \sqrt{(1 + \omega^2 C^2 R^2)} \cdot \sin(\omega t + \phi) \quad [20.13]$$

$$\text{where } \tan \phi = \frac{\omega C V_m}{V_m/R} = \omega C R$$

If $i = 0$, i.e. diode cuts off, when $t = t_2$ then

$$\sin(\omega t_2 + \phi) = 0$$

$$\therefore \omega t_2 + \phi = \pi$$

$$\therefore \omega t_2 = \pi - \phi \quad [20.14]$$

If $\omega C R \gg 1$ then

$$\phi \approx \frac{\pi}{2}$$

$$\text{and } \omega t_2 \approx \pi - \frac{\pi}{2} = \frac{\pi}{2}$$

i.e. the diode ceases to conduct near the instant at which v has its positive maximum value.

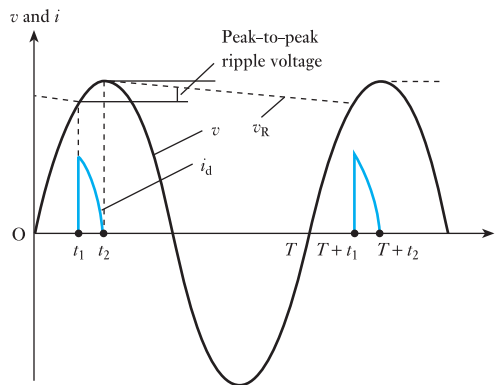
While the diode is non-conducting, C will discharge through R and the load voltage will be given by

$$v_R = V e^{-\frac{(t-t_2)}{CR}}$$

where V is the capacitor voltage at the instant the diode cuts off. This will equal approximately the peak value of the supply voltage if $\omega CR \gg 1$. The diode will start to conduct again during the period that the supply voltage is positive and increasing. This instant can be determined by equating $V e^{-(t-t_2)/CR}$ and $V_m \sin \omega t$, one solution of which will give $T + t_1$, where T is the period of the supply voltage and t_1 the instant of time at which the diode starts to conduct.

In order to keep the variation in load voltage down during the period when the diode is non-conducting, a long time constant CR compared with the period of the supply voltage is required. Note, however, from relation [20.13] that the peak value of the diode current increases with C and therefore care must be taken to ensure that the maximum allowable diode peak current is not exceeded. Waveforms for the circuit are shown in Fig. 20.9.

Fig. 20.9 Waveforms for half-wave rectifier with capacitor input filter



A similar analysis could be carried out for the full-wave circuit. The operation is identical to that of the half-wave circuit during the charging period, but the capacitor discharges into the load resistance for a shorter period, giving less amplitude of ripple for a given time constant during the non-conducting period. Figure 20.10 shows waveforms for a full-wave circuit with a capacitor input filter.

Fig. 20.10 Waveforms for full-wave rectifier with capacitor input filter

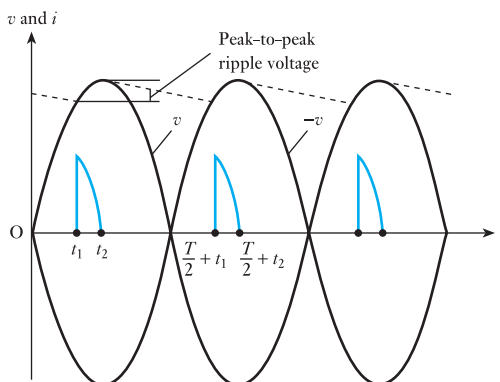


Fig. 20.11 Smoothing circuits.

- (a) Series inductor filter;
 (b) LC filter; (c) π filter;
 (d) resistive filter

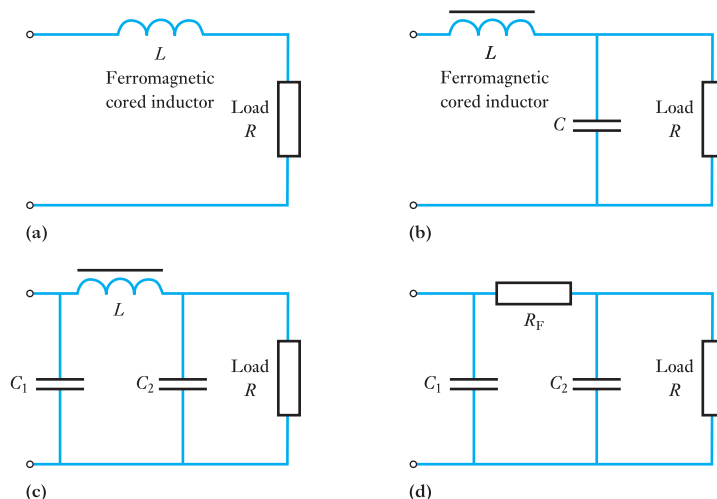


Figure 20.11 shows some other types of filter circuits used in practice for smoothing purposes. The ideal arrangement is to connect a low-valued impedance at the frequency being used and which presents a high resistance to direct current, e.g. a capacitor, in parallel with the load, and/or a high-valued impedance at the frequency being used and which presents a low resistance to direct current, e.g. an inductor, in series with the load. This has the effect of minimizing the ripple across the load while having little effect on the direct voltage developed at the input to the filter circuit. The circuit of Fig. 20.11(d) does not meet these requirements fully in so much as the resistor R will reduce the direct voltage as well as the ripple. The circuit is useful, however, in low current applications where the direct voltage drop across R can be kept small; a resistor has an economic advantage over a ferromagnetic-cored inductor.

20.6

Zener diode

If the reverse voltage across a p–n junction is gradually increased, a point is reached where the energy of the current carriers is sufficient to dislodge additional carriers. These carriers, in turn, dislodge more carriers and the junction goes into a form of avalanche breakdown characterized by a rapid increase in current as shown in Fig. 19.18. The power due to a relatively large reverse current, if maintained for an appreciable time, can easily ruin the device.

Special junction diodes, often known as *Zener diodes*, but more appropriately termed *voltage regulator diodes*, are available in which the reverse breakdown voltage is in the range of about 4 V to about 75 V, the actual voltage depending upon the type of diode.

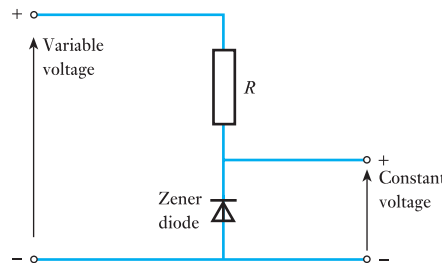
When the voltage regulator diode is forward biased, it behaves as a normal diode. With a small reverse voltage, the current is the sum of the surface leakage current and the normal saturation current due to the thermally generated holes and electrons (section 19.6). This current is only a few microamperes; but as the reverse voltage is increased, a value is reached at

which the current suddenly increases very rapidly (Fig. 19.18). As already mentioned, this is due to the increased velocity of the carriers being sufficient to cause ionization. The carriers resulting from ionization by collision are responsible for further collisions and thus produce still more carriers. Consequently the numbers of carriers, and therefore the current, increase rapidly due to this avalanche effect.

The voltage across the regulator diode after breakdown is termed the *reference voltage* and its value for a given diode remains practically constant over a wide range of current, provided the maximum permissible junction temperature is not exceeded.

Figure 20.12 shows how a voltage regulator diode (or Zener diode) can be used as a voltage stabilizer to provide a constant voltage from a source whose voltage may vary appreciably. A resistor R is necessary to limit the reverse current through the diode to a safe value.

Fig. 20.12 A voltage stabilizer



Summary of important formulae

For a half-wave rectifier,

$$I_{dc} = \frac{1}{\pi} I_m = 0.318 I_m \quad [20.2]$$

$$I_{rms} = \frac{1}{2} I_m = 0.5 I_m \quad [20.3]$$

$$V_{dc} = \frac{1}{\pi} V_m = 0.318 V_m \quad [20.4]$$

$$V_{rms} = \frac{1}{2} V_m = 0.5 V_m \quad [20.5]$$

For a full-wave rectifier,

$$I_{dc} = \frac{2}{\pi} I_m = 0.637 I_m \quad [20.7]$$

$$I_{rms} = \frac{1}{\sqrt{2}} I_m = 0.707 I_m \quad [20.8]$$

$$V_{dc} = \frac{2}{\pi} V_m = 0.637 V_m \quad [20.9]$$

$$V_{rms} = \frac{1}{\sqrt{2}} V_m = 0.707 V_m \quad [20.10]$$

21.1 Introduction

Having become familiar with the semiconductor diode in Chapter 19, it is now possible to progress to an understanding of the transistor, which is fundamental to most amplifier arrangements, as well as many switching arrangements associated with digital systems.

There are two basic types of transistor: (1) the bipolar junction transistor; (2) the field effect transistor (FET).

Early transistor circuits depended entirely on the junction transistor and for that reason it will be considered first. However its large size has reduced its range of applications in practice.

21.2 Bipolar junction transistor

A bipolar junction transistor is a combination of two junction diodes and consists of either a thin layer of p-type semiconductor sandwiched between two n-type semiconductors, as in Fig. 21.1(a), and referred to as an n-p-n transistor, or a thin layer of an n-type semiconductor sandwiched between two p-type semiconductors, as in Fig. 21.1(b), and referred to as a p-n-p transistor. The thickness of the central layer, known as the *base*, is of the order of $25\ \mu\text{m}$ (or $25 \times 10^{-6}\ \text{m}$).

The junction diode formed by n_1 -p in Fig. 21.1(a) is biased in the forward direction by a battery B_1 so that free electrons are urged from n_1 towards p. Hence n_1 is termed an *emitter*. On the other hand, the junction diode formed by n_2 -p in Fig. 21.1(a) is biased in the reverse direction by battery B_2 so that if battery B_1 were disconnected, i.e. with zero emitter current, no current would flow between n_2 and p apart from that due to the thermally generated minority carriers referred to in section 19.6. However, with B_1 connected as in Fig. 21.1(a), the electrons from emitter n_1 enter p and diffuse through the base until they come within the influence of n_2 , which is connected to the positive terminal of battery B_2 . Consequently the electrons which reach n_2 are collected by the metal electrode attached to n_2 ; hence n_2 is termed a *collector*.

Fig. 21.1 Arrangement of a bipolar junction transistor.

(a) n-p-n; (b) p-n-p

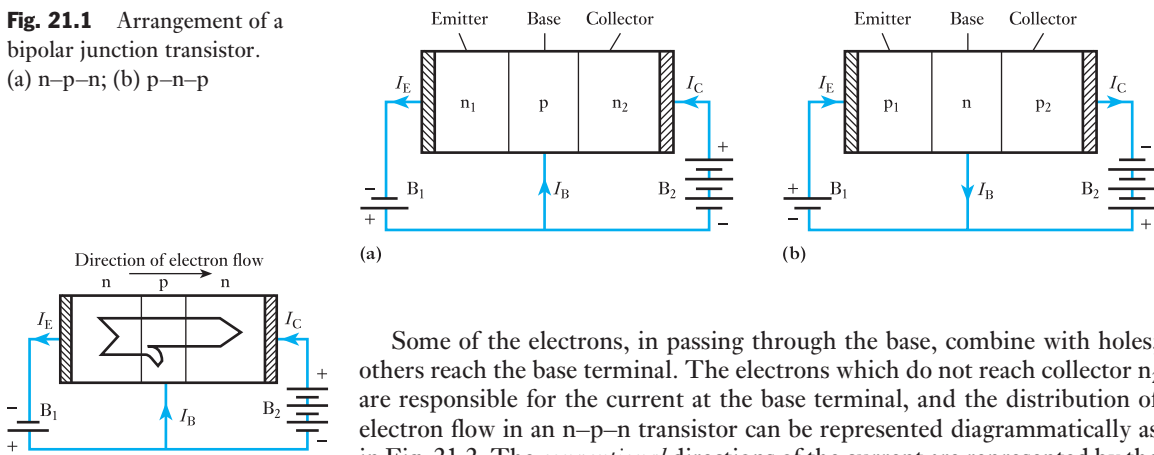


Fig. 21.2 Electron flow in an n-p-n transistor

Some of the electrons, in passing through the base, combine with holes; others reach the base terminal. The electrons which do not reach collector n_2 are responsible for the current at the base terminal, and the distribution of electron flow in an n-p-n transistor can be represented diagrammatically as in Fig. 21.2. The *conventional* directions of the current are represented by the arrows marked I_E , I_B and I_C , where I_E is the emitter current, I_B the base current and I_C the collector current.

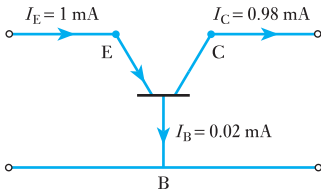


Fig. 21.3 Currents in emitter, collector and base circuits of a p-n-p transistor

By making the thickness of the base very small and the impurity concentration in the base much less than in the emitter and collector, the free electrons emerging from the emitter have little opportunity of combining with holes in the base, with the result that about 98 per cent of these electrons reach the collector.

By Kirchhoff's first law, $I_E = I_B + I_C$, so that if $I_C = 0.98I_E$, then $I_B = 0.02I_E$; thus, when $I_E = 1$ mA, $I_C = 0.98$ mA and $I_B = 0.02$ mA, as in Fig. 21.3.

In the above explanation we have dealt with the n-p-n transistor, but exactly the same explanation applies to the p-n-p transistor of Fig. 21.1(b) except that the movement of electrons is replaced by the movement of holes.

21.3

Construction of a bipolar transistor

The first step is to purify the silicon or germanium so that any impurity does not exceed about 1 part in 10^{10} . Various methods have been developed for attaining this exceptional degree of purity, and intensive research is still being carried out to develop new methods of purifying and of doping silicon and germanium.

In one form of construction of the p-n-p transistor, the purified material is grown as a single crystal, and while the material is in a molten state, an n-type impurity (e.g. antimony in the case of germanium) is added in the proportion of about 1 part in 10^8 . The solidified crystal is then sawn into slices about 0.1 mm thick. Each slice is used to form the base region of a transistor; thus in the case of n-type germanium, a pellet of p-type impurity such as indium is placed on each side of the slice, the one which is to form the collector being about three times the size of that forming the emitter. One reason for the larger size of the collector bead is that the current carriers from the emitter spread outwards as they pass through the base, and the larger area of the collector enables the latter to collect these carriers more effectively. Another reason is that the larger area assists in dissipating the greater power loss at the collector-base junction. This greater loss is due to the p.d. between collector and base being greater than that between emitter and base.

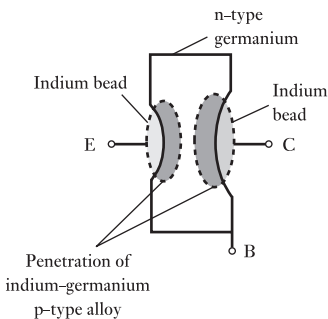


Fig. 21.4 Construction of a p-n-p germanium transistor

The assembly is heated in a hydrogen atmosphere until the pellets melt and dissolve some of the germanium from the slice, as shown in Fig. 21.4. Leads for the emitter and collector are soldered to the surplus material in the pellets to make non-rectifying contacts, and a nickel tab is soldered to make connection to the base. The assembly is then hermetically sealed in a metal or glass container, the glass being coated with opaque paint. The transistor is thus protected from moisture and light.

21.4

Common-base and common-emitter circuits

In Fig. 21.1 the transistor is shown with the base connected directly to both the emitter and collector circuits; hence the arrangement is referred to as a *common-base* circuit. The conventional way of representing this arrangement is shown in Fig. 21.5(a) and (b) for n-p-n and p-n-p transistors respectively, where E, B and C represent the emitter, base and collector terminals. The arrowhead on the line joining the emitter terminal to the base indicates the conventional direction of the current in that part of the circuit.

Fig. 21.5 Common-base circuits. (a) n-p-n; (b) p-n-p

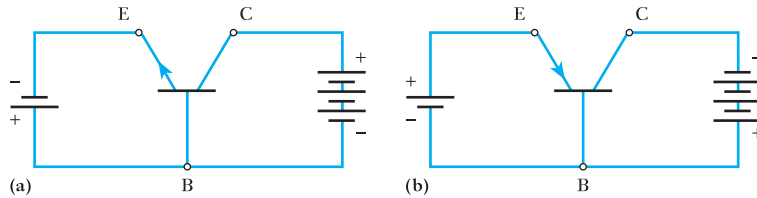


Fig. 21.6 Common-emitter circuits. (a) n-p-n; (b) p-n-p

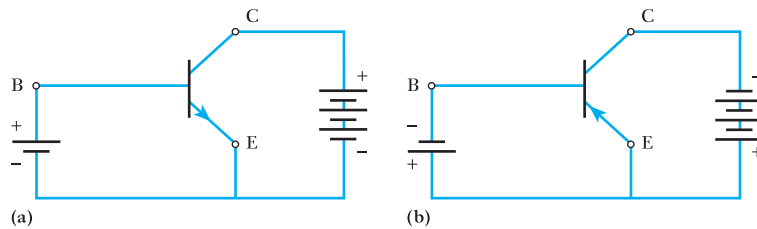


Figure 21.6 shows the *common-emitter* method of connecting a transistor. In these diagrams it will be seen that the emitter is connected directly to the base and collector circuits. This method is more commonly used than the common-base circuit owing to its higher input resistance and the higher current and power gains.

The common-collector circuit is used only in special cases, and will therefore not be considered here.

21.5 Static characteristics for a common-base circuit

Figure 21.7 shows an arrangement for determining the static characteristics of an n-p-n transistor used in a common-base circuit. The procedure is to maintain the value of the emitter current, indicated by A_1 , at a constant value, say 1 mA, by means of the slider on R_1 , and note the readings on A_2 for various values of the collector-base voltage given by voltmeter V_2 . The

Fig. 21.7 Determination of static characteristics for a common-base n-p-n transistor circuit

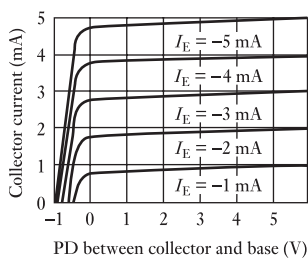
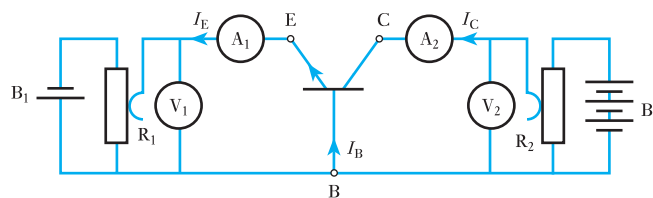


Fig. 21.8 Static characteristics for a common-base n-p-n transistor circuit

test is repeated for various values of the emitter current and the results are plotted as in Fig. 21.8.

In accordance with BS 3363, the current is assumed to be positive when its direction is from the external circuit towards the transistor terminal, and the voltage V_{CB} is positive when C is positive relative to B. Hence, for an n-p-n transistor, the collector current and collector-base voltage are positive but the emitter current is negative. For a p-n-p transistor, all the signs have to be reversed.

From Fig. 21.8 it will be seen that for positive values of the collector-base voltage, the collector current remains almost constant, i.e. nearly all the electrons entering the base of an n-p-n transistor are attracted to the

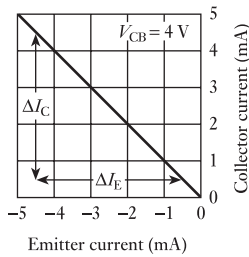


Fig. 21.9 Relationship between collector and emitter currents for a given collector–base voltage

collector. Also, for a given collector–base voltage, the collector current is practically proportional to the emitter current. This relationship is shown in Fig. 21.9 for V_{CB} equal to 4 V. The ratio of the change, ΔI_C , of the collector current to the change, ΔI_E , of the emitter current (neglecting signs), for a given collector–base voltage, is termed the *current amplification factor for a common-base circuit* and is represented by the symbol α , i.e.

$$\alpha = \frac{\Delta I_C}{\Delta I_E} \quad \text{for a given value of } V_{CB} \quad [21.1]$$

= slope (neglecting signs) of $\frac{I_C}{I_E}$ graph in Fig. 21.9

21.6

Static characteristics for a common-emitter circuit

Figure 21.10 shows an arrangement for determining the static characteristics of an n–p–n transistor used in a common-emitter circuit. Again, the procedure is to maintain the *base* current, I_B , through a microammeter A_1 constant at, say, $25 \mu\text{A}$, and to note the collector current, I_C , for various values of the collector–emitter voltage V_{CE} , the test being repeated for several values of the base current, and the results are plotted as shown in Fig. 21.11. For a given voltage between collector and emitter, e.g. for $V_{CE} = 4 \text{ V}$, the relationship between the collector and base currents is practically linear as shown in Fig. 21.12.

Fig. 21.10 Determination of static characteristics for a common-emitter n–p–n transistor circuit

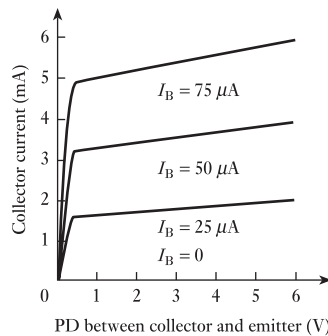
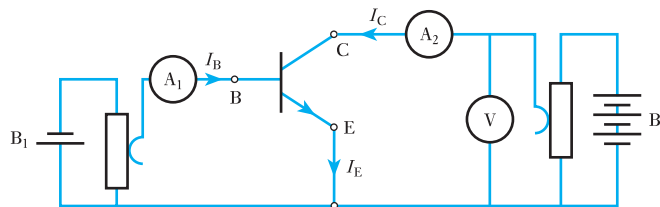


Fig. 21.11 Static characteristics for a common-emitter n–p–n transistor circuit

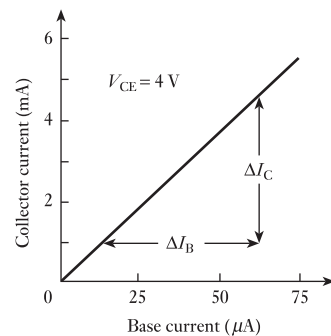


Fig. 21.12 Relationship between collector and base currents, for a given collector–emitter voltage

The ratio of the change, ΔI_C , of the collector current to the change, ΔI_B , of the base current, for a given collector–emitter voltage, is termed the *current amplification factor for a common-emitter circuit* and is represented by the symbol β , i.e.

$$\beta = \frac{\Delta I_C}{\Delta I_B} \quad \text{for a given value of } V_{CE} \quad [21.2]$$

= slope of graph in Fig. 21.12

21.7

Relationship between α and β

From Figs 21.7 and 21.10 it is seen that

$$I_E = I_C + I_B$$

$$\therefore \Delta I_E = \Delta I_C + \Delta I_B$$

From expression [21.1]

$$\begin{aligned} \alpha &= \frac{\Delta I_C}{\Delta I_E} \\ &= \frac{\Delta I_C}{(\Delta I_C + \Delta I_B)} \end{aligned}$$

$$\begin{aligned} \therefore \frac{1}{\alpha} &= 1 + \frac{\Delta I_B}{\Delta I_C} \\ &= 1 + \frac{1}{\beta} = \frac{(1 + \beta)}{\beta} \end{aligned}$$

Hence

$$\alpha = \frac{\beta}{(1 + \beta)} \quad [21.3]$$

and

$$\beta = \frac{\alpha}{(1 - \alpha)} \quad [21.4]$$

Thus, if

$$\alpha = 0.98, \quad \beta = 0.98/0.02 = 49$$

and if

$$\alpha = 0.99, \quad \beta = 0.99/0.01 = 99$$

i.e. a small variation in α corresponds to a large variation in β . It is therefore better to determine β experimentally and calculate therefrom the corresponding value of α by means of expression [21.3].

A transistor in a common-emitter circuit has the base–emitter junction forward biased and the collector–base junction reverse biased. Under these conditions and in the absence of an input signal

$$I_E = I_B + I_C$$

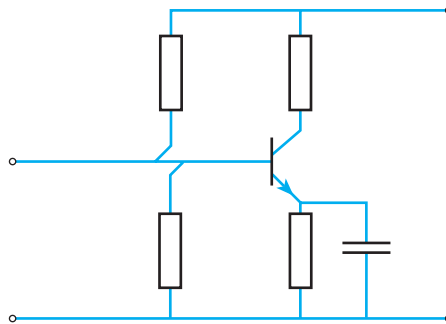
If the emitter is not connected, $I_E = 0$. However, we find there is still a small collector current – this is a leakage current crossing the reverse-biased collector–base junction. This leakage current is given the symbol I_{CB0} being the collector–base current with zero emitter current.

Let us now reconnect the emitter. The leakage current is still present and therefore the total collector current is given by

$$\begin{aligned} I_C &= \alpha I_E + I_{CB0} \\ &= \alpha(I_B + I_C) + I_{CB0} \\ I_C(1 - \alpha) &= \alpha I_B + I_{CB0} \\ I_C &= \frac{\alpha}{1 - \alpha} I_B + \frac{I_{CB0}}{1 - \alpha} \\ I_C &= \beta I_B + \frac{I_{CB0}}{1 - \alpha} \end{aligned}$$

If I_{CB0} is very small, the last term of this expression can be neglected and the relation reduces to that of equation [21.2]. The leakage current is not necessarily negligible. The leakage current is temperature dependent, hence an increase in temperature causes the leakage current to rise. In turn this results in an increase in collector current and a change in bias conditions. The change in bias conditions is stabilized by using a potential divider, as shown in Fig. 21.13. This holds the base voltage almost constant. It is also necessary to introduce a resistor into the emitter connection.

Fig. 21.13 Bias point stabilization



21.8 Load line for a transistor

Let us consider an n–p–n transistor used in a common-base circuit (Fig. 21.14), together with an a.c. source S having an internal resistance R_S , a load resistance R and bias batteries B_1 and B_2 giving the required transistor currents.

Let us assume that the voltage between the collector and base is to be 3 V when there is *no alternating voltage* applied to the emitter, and that, with an emitter current of 3 mA, the corresponding collector current is 2.9 mA. This corresponds to point D in Fig. 21.15. Also suppose the resistance R of the load to be $1000\ \Omega$. Consequently the corresponding p.d. across R is 0.0029×1000 , namely 2.9 V, and the total bias supplied by battery B_2 must be $3 + 2.9$, namely 5.9 V. This is represented by point P in Fig. 21.15.

Fig. 21.14 An n-p-n transistor with load resistance R

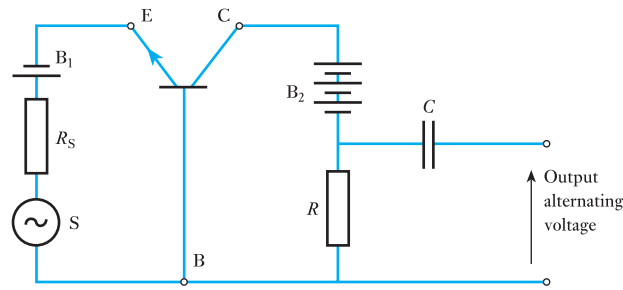
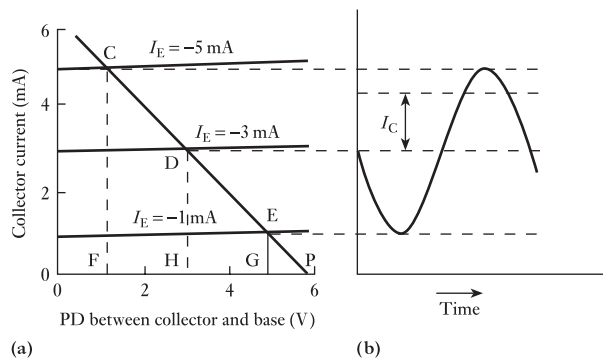


Fig. 21.15 Graphical determination of the output current of a transistor in a common-base circuit



The straight line PD drawn through P and D is the *load line*, the inverse of the slope of which is equal to the resistance of the load, e.g. $HP/DH = 2.9/0.0029 = 1000 \Omega$. The load line is the locus of the variation of the collector current for any variation in the emitter current. Thus, if the emitter current decreases to 1 mA, the collector current decreases to GE and the p.d. across R decreases to GP. On the other hand, if the emitter current increases to 5 mA, the collector current increases to FC and the p.d. across R increases to FP. Hence, if the emitter current varies sinusoidally between 1 and 5 mA, the collector current varies as shown at (b) in Fig. 21.15. This curve will be sinusoidal if the graphs are linear and are equally spaced over the working range.

The function of capacitor C is to eliminate the d.c. component of the voltage across R from the output voltage.

If I_c is the r.m.s. value of the *alternating* component of the collector current, $I_c R$ gives the r.m.s. value of the alternating component of the output voltage and $I_c^2 R$ gives the output power due to the alternating e.m.f. generated in S . Note that a lower-case subscript indicates an r.m.s. value compared with a capital for the d.c. value – the difference requires good eyesight! Hence the larger the value of R , the greater the voltage and power gains, the maximum value of R for a given collector supply voltage being limited by the maximum permissible distortion of the output voltage.

A similar procedure can be used to determine the output voltage and power for a transistor used in a common-emitter circuit.